MANIPAL ACADEMY of HIGHER EDUCATION

Question Paper - Report

10-Oct-2024 14:35:21
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Question Paper

MANIPAL ACADEMY OF HIGHER EDUCATION

ENGINEERING MATHEMATICS - III [MAT 2126]

Marks: 30 **Duration: 90 mins.**

A

Answer all the questions. Section Duration: 20 mins

1) Let G be a simple graph with 9 vertices and 15 edges. If the degree of 8 of its vertices are (2,2,3,3,4,4,5,5), then the degree of the 9th vertex is _____. (0.5)

2 6 1 4

2) Which of the following binary operations is associative? (0.5)

The operation * on the set $\mathbb{Z}$ of integers defined by $a*b = a - b$.

The operation * on the set $\mathbb{Q}$ of rational numbers defined by $a * b = \frac{a+b}{5}$

The operation * on the set $\mathbb{Q}$-{0} of non-zero rational numbers defined by $a * b = \frac{a}{b}$.

The operation * on the set $\mathbb{R}$ of real numbers defined by a * b = a + b + ab.

3) Consider the following statements (0.5)

P : Every complete graph with *n* vertices is regular with regularity (*n* - 1).

Q: Every cycle graph with *n* vertices is regular with regularity 2.

Only *P* is true Only *Q* is true Both *P* and *Q* are true Neither *P* nor *Q* is true

4) If a simple graph *G* has 10 vertices and 5 edges, then the number of edges in $\bar{G}$ is _____. (0.5)

5 15 40 45

5) The number of distinct left cosets of the subgroup (5$\mathbb{Z}$, +) of ($\mathbb{Z}$,+) (where 5$\mathbb{Z}$ denotes the set of all multiples of 5) is _____. (0.5)

5 4 6 3

6) If *G* is a finite group and the number of elements in *G* is 30, then what is the possible order of a subgroup *H* of *G*? (0.5)

4 7 8 15

7) What is the order of the element 2 in the group ($\mathbb{Z}_6$, $+_6$) (where $+_6$ denotes addition modulo 6)? (0.5)

6 3 2 1

8) Which of the following statements is FALSE?

Any subgroup of an Abelian group is Abelian | Any subgroup of a cyclic group is Abelian | Any finite group is Abelian. | There exists an Abelian group with a non-cyclic subgroup. (0.5)

9) The determinant of the adjacency matrix of a cycle on 4 vertices is _____.

(0.5)

0 1 -1 2

10) The diameter of the cycle graph with 6 vertices is _____.

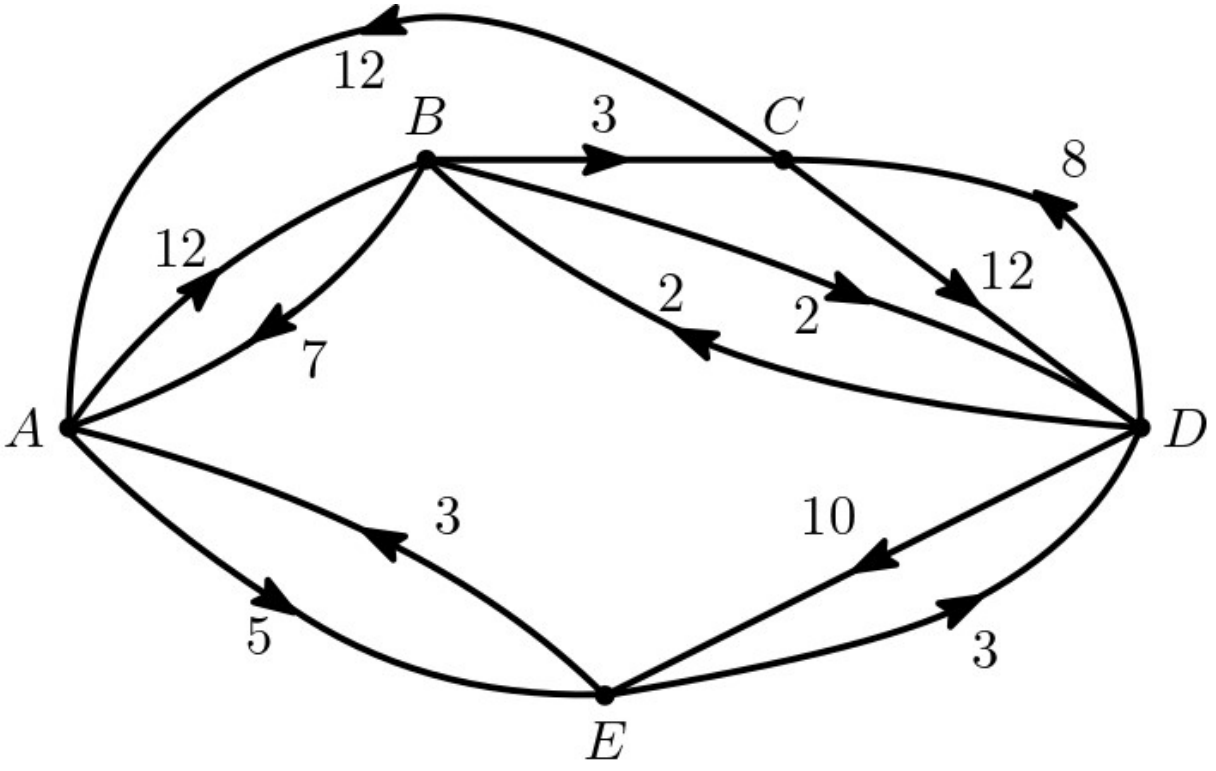
(0.5)

3 2 1 6

B

Answer all the questions.

11) Using Dijkstra's algorithm, find shortest paths from A to all the other vertices in the graph given below.



(4)

12) Show that every subgroup of a cyclic group is cyclic. (3)

13) Draw 3 non-isomorphic simple graphs on 4 vertices with exactly 3 edges, giving brief reasons why they are non-isomorphic. (3)

14) Let G be a group and H a subgroup of G. Show that any two left cosets of H in G are either disjoint or identical. Also illustrate this result with an example. (3)

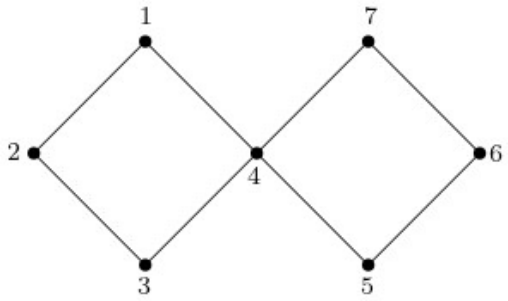
15) Show that a graph G is a tree if and only if between every pair of vertices there exist a unique path. Draw a tree on 5 vertices. (3)

16) Show that a group G is Abelian if and only if, for all $a, b \in G$, $(ab)^2 = a^2b^2$. Give an example of an Abelian group. (3)

17) Let G be a self-complementary graph. Show that the number of vertices in G is of the form $4k$ or $4k + 1$. (2)

18) Let G be a finite group and let H and K be subgroups with relatively prime order (i.e. the gcd of their orders is 1). Then show that $H \cap K = \{e\}$. (2)

19) Show that the graph given below is bipartite, by listing the vertices in each of the partite sets V_1 and V_2. State, with justification, whether this graph is Eulerian. (2)



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